

UNIVERSITY OF ENGINEERING AND TECHNOLOGY LAHORE



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Task 1

```
using System;
```

```
namespace CSharpDev
```

```
{
```

```
    class Student
```

```
    {
```

```
        public string name;
```

```
        public string session;
```

```
        public bool isDayScholar;
```

```
        public int EntryTestMarks;
```

```
        public int HSMarks;
```

```
        public double
```

```
        CalculateMerit()
```

```
        {
```

```
            return (HSMarks * 0.5) +  
            EntryTestMarks;
```

```
        }
```

```
    }
```

```
    class Hostelite : Student
```

```
    {
```

```
        public int RoomNumber;
```

```
        public bool  
        isFridgeAvailable;
```

```

        public bool
isInternetAvailable;

        private int hostelFee =
3000;

        public int GetHostelFee()
        {
            return hostelFee;
        }
    }

class DayScholar : Student
{
    public string pickupPoint;
    public int BusNumber;
    public int pickupDistance;

    public int GetBusFee()
    {
        int fee = pickupDistance *
100;
        return fee;
    }
}

class Program
{

```

```
static void Main(string[]
args)
{
    DayScholar ds = new
DayScholar();

    ds.name = "Ahmad";

    ds.BusNumber = 1;

Console.WriteLine(std.name + "
is Allocated Bus " +
std.BusNumber);

    Hostelite h = new
Hostelite();

    h.name = "Ali";

    h.RoomNumber = 12;

Console.WriteLine(std.name + "
is Allocated Bus " +
std.RoomNumber);

    Console.ReadKey();

}

}
```

Output :

```
ahmad Merit: 545.5
ahmad is allocated bus 1

=== Code Execution Successful ===
```

Self Assessment 1 :

using System;

class Ladder

{

 public double length;

 public string colour;

 public Ladder(double length,
 string colour)

 {

 this.length = length;

 this.colour = colour;

 }

}

class HosePipe

{

 public string material;

 public string shape;

 public double diameter;

 public double waterFlowRate;

 public HosePipe(string
 material, string shape, double
 diameter, double
 waterFlowRate)

```
{  
    this.material = material;  
    this.shape = shape;  
    this.diameter = diameter;  
    this.waterFlowRate =  
waterFlowRate;  
}  
}
```

```
class FireFighter
```

```
{  
    public string name;  
  
    public FireFighter(string  
name)  
    {  
        this.name = name;  
    }  
  
    public virtual void drive()  
    {  
        Console.WriteLine(name + "  
is driving the fire truck.");  
    }  
  
    public virtual void extinguish()  
    {
```

```
        Console.WriteLine(name + "
is extinguishing the fire.");
    }
}
```

```
class FireChief : FireFighter
{
    public FireChief(string name) :
base(name) { }
```

```
    public override void drive()
    {
        Console.WriteLine("Fire
Chief " + name + " is driving the
truck.");
    }
```

```
    public override void
extinguish()
    {
        Console.WriteLine("Fire
Chief " + name + " is putting out
the fire.");
    }
}
```

```
class FireTruck
{
    private Ladder ladder;
```

```
public HosePipe hosePipe;
```

```
public FireFighter driver;
```

```
public FireTruck(Ladder  
ladder, HosePipe hosePipe,  
FireFighter driver)
```

```
{
```

```
    this.ladder = ladder;
```

```
    this.hosePipe = hosePipe;
```

```
    this.driver = driver;
```

```
}
```

```
public void displayInfo()
```

```
{
```

```
    Console.WriteLine("////Fire  
Truck Info////");
```

```
    Console.WriteLine("Driver  
: " + driver.name);
```

```
    Console.WriteLine("Ladder  
: " + ladder.length + "m,  
colour=" + ladder.colour);
```

```
    Console.WriteLine("Hose  
Pipe : " + hosePipe.material + "  
" + hosePipe.shape
```

```
        + ", diameter = " +  
hosePipe.diameter
```



```
        + ", flow = " +  
hosePipe.waterFlowRate +  
"L/s");  
    }  
}
```

```
class Program  
{  
    static void Main(string[] args)  
    {  
        FireFighter ff1 = new  
FireFighter("Usman");  
  
        FireChief chief = new  
FireChief("Hassan");  
  
        Ladder ladder = new  
Ladder(10.5, "Red");  
  
        HosePipe hose = new  
HosePipe("synthetic rubber",  
"cylindrical", 0.08, 3.5);  
  
        FireTruck truck = new  
FireTruck(ladder, hose, ff1);  
  
        truck.displayInfo();  
  
        Console.WriteLine();  
  
        ff1.drive();  
  
        ff1.extinguish();  
  
        Console.WriteLine();  
    }  
}
```

```

        chief.drive();

        chief.extinguish();

        Console.ReadKey();
    }
}

```

Output :

```

////Fire Truck Info////
Driver      : Usman
Ladder      : 10.5m, colour=Red
Hose Pipe   : synthetic rubber, cylindrical, diameter = 0.08, flow = 3.5L/s

Usman is driving the fire truck.
Usman is extinguishing the fire.

Fire Chief Hassan is driving the truck.
Fire Chief Hassan is putting out the fire.

```

Self Assessment 2 :

```

using System;

class Bicycle
{
    private int cadence;
    private int gear;
    private int speed;
}

```

```
    public Bicycle(int cadence, int  
speed, int gear)
```

```
{  
    this.cadence = cadence;  
    this.speed = speed;  
    this.gear = gear;  
}
```

```
    public void setCadence(int  
cadence)
```

```
{  
    this.cadence = cadence;  
}
```

```
    public void setGear(int gear)
```

```
{  
    this.gear = gear;  
}
```

```
    public void applyBrake(int  
decrement)
```

```
{  
    speed -= decrement;  
    if (speed < 0) speed = 0;  
}
```

```
    public void speedUp(int  
increment)
```

```
    {  
        speed += increment;  
    }
```

```
    public void displayInfo()
```

```
    {
```

```
        Console.WriteLine("Cadence: " +  
cadence
```

```
            + " Gear: " + gear
```

```
            + " Speed: " +  
speed);
```

```
    }
```

```
}
```

```
class MountainBike : Bicycle
```

```
{
```

```
    private int seatHeight;
```

```
    public MountainBike(int  
seatHeight, int cadence, int  
speed, int gear)
```

```
    {  
        : base(cadence, speed,  
gear)
```

```
    {
```

```
        this.seatHeight =  
seatHeight;
```

```
}
```

```
    public void setSeatHeight(int  
seatHeight)
```

```
{
```

```
    this.seatHeight =  
seatHeight;
```

```
}
```

```
    public void  
displayMountainBikeInfo()
```

```
{
```

```
    base.displayInfo();
```

```
    Console.WriteLine("Seat  
Height: " + seatHeight);
```

```
}
```

```
}
```

```
class Program
```

```
{
```

```
    static void Main(string[] args)
```

```
{
```

```
        MountainBike mb = new  
MountainBike(30, 80, 0, 1);
```

```
        mb.speedUp(20);
```

```
        mb.setGear(3);
```

```
        mb.setCadence(90);
```

```
mb.setSeatHeight(35);
```

```
mb.displayMountainBikeInfo();
```

```
mb.applyBrake(5);
```

```
mb.displayMountainBikeInfo();
```

```
}
```

```
}
```

Output :

```
Cadence: 90 Gear: 3 Speed: 20  
Seat Height: 35  
Cadence: 90 Gear: 3 Speed: 15  
Seat Height: 35  
PS C:\Users\NoumanPC\source\repos>
```

